From echolocation clicks to population counts - Harbour porpoise (*Phocoena phocoena*) density in the North Sea

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How to convert click detections into density with varying behaviour, environment and detector performance?

Are clicks emitted while **facing** the sensor **within** the area of detection? How many/loud?

Is the click **distorted** during emission/propagation?

Amplitude declines with distance

Is the received click **classified** correctly?

Is the **amplitude** high enough for **detection** after absorption into water, relative to noise and detector threshold?

Click!

Waves, currents

Sediment

Weather

Cetaceans /Animals

Propeller cavitation

Boat sonar

Porpoise habitat ecology factors: Prey distribution, depth, light, weather, waves, reverberation level

Environment

Noise

Signal-to-noise ratio: Signal (click) amplitude relative to noise levels

Noise and false positives

Behaviour

Source levels: Sound intensity at 1m distance in the direction of maximum intensity (on-axis); behaviourally adjusted (e.g. reduced for prey capture & where reverberation high)

Click rate depends on behavioural state (high during foraging) environmental complexity, prey distribution

Frequency: Clicks **detectable until 500 - 1000 m** from sensor; noise levels are relatively low.

Click production

Click projection

Movement and orientation relative to sensor depend on behavioural state (e.g. scanning prey search, rolling capture)

Distorted off-axis clicks: lower amplitude, broadband frequency: increases detectability close to sensor, depends on detection threshold

Narrow sound beam like a flashlight compressed in shallow waters; behaviourally modified (e.g. widened during prey capture, narrowed if complexity high)

Sound propagation

Environmental complexity: Substrate, structure, flora

Water depth affects sound attenuation and bottom/surface reflections in shallow water. **Salinity, temperature, pressure** minor effects here

Weather, currents

Acoustic Sensor

Variation between detection performance and sensor **deterioration**

Study design: Sensor number, density, geometry, depth; Arrays estimate distance

Detection threshold: How much louder than noise level for a click to be detected?

False positive chance trains vs Missed detections trade-off: Analysis on click-by-click or click-sequence basis?

How to infer group size from click detections?

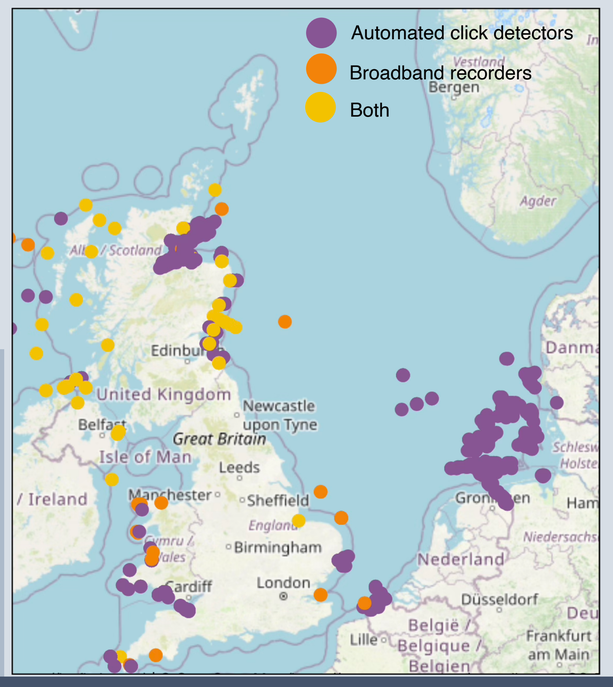
Detection and classification performance

Environment, behaviour and detector performance vary over space and time – Needs to be reflected in the variables of the density equation:

- **Estimate the probability that a porpoise available in the area is actually detected**
- **Estimate the area in which porpoises are detected reliably**
- **Convert click detection unit into a number of animals**
- **Estimate the proportion of detections that are falsely classed as porpoise clicks**

$$Porpoise\ Density = \frac{Detections * (1 - FalsePositiveRate)}{DetectionProbability * Area * Multiplier}$$

Click detections can be expressed in different units:
- **Encounter rates & duration, Waiting Time (Time between encounters)**
- **Detection positive seconds/minutes/hours/days (Time unit with at least 1 detection)**
- **Clicks per detection positive time unit (Because time units saturate with 1 detection)**



Map: **Chloe Malinka et al.**

DTO-BioFlow
Integration of biodiversity monitoring data into the Digital Twin Ocean

Visit the website: dto-bioflow.eu

- PhD part of project funded by European Commission: DTO-BioFlow
- Aims to consolidate, process, standardize ocean biodiversity data (often inaccessible and unstandardized)
- Use genomic, imaging, biologging, passive acoustic and citizen science data to inform the EU Digital Twin Ocean model, a virtual representation of the marine environment to inform conservation efforts and make predictions under different scenarios

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